

A Regression-Based Mathematical Framework for Predicting Battery Life Degradation in Electric Vehicles

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Abstract—The performance and reliability of electric vehicles are closely tied to the health and lifespan of their batteries. Over time, batteries experience gradual degradation due to repeated charging cycles, temperature variations, and operational stress, leading to reduced capacity and efficiency. This paper presents a regression-based mathematical framework designed to predict the life degradation of electric vehicle batteries using a purely data-driven and software-oriented approach. The proposed model integrates regression analysis with an entropy-based evaluation of gradient stability, offering a mathematically interpretable method to estimate degradation trends. By analysing the relationship between key influencing parameters such as charge cycles, temperature, voltage, and discharge capacity, the model establishes a predictive relationship that reflects real-world performance decline. The introduction of entropy in gradient assessment provides additional insight into model convergence and stability, ensuring reliable predictive accuracy. The framework is designed for simplicity, reproducibility, and computational efficiency, making it suitable for implementation on standard systems without specialized hardware. Experimental validation through open-access datasets demonstrates the model's effectiveness in identifying degradation behaviour and forecasting remaining battery life. The proposed study contributes to the advancement of intelligent battery health monitoring and presents a pathway toward sustainable and cost-effective electric vehicle maintenance through interpretable machine learning techniques. This paper proposes a regression-based mathematical framework for modeling battery degradation using features such as charge throughput, temperature variation, and resistance growth. Experimental evaluation on the NASA battery dataset demonstrates improved prediction accuracy, achieving an R^2 score of 0.9951 and reduced RMSE compared to baseline models. The proposed model outperforms baseline methods including SVR and Random Forest in terms of prediction accuracy.

Index Terms—Battery Life Prediction, Regression-Based Machine Learning, Entropy Gradient Analysis, Electric Vehicle Degradation, Mathematical Modelling, Predictive Maintenance, Data-Driven Energy Systems

I. INTRODUCTION

The global move towards electric vehicles (EVs) marks an important step in addressing environmental issues and reaching sustainable mobility goals. Central to this shift is the lithium-

ion battery, a complex and costly part whose performance directly affects the EV's range, lifespan, and overall reliability [7]. However, the long-term health of these batteries suffers from capacity degradation, a complicated process driven by repeated charge-discharge cycles, temperature changes, and internal electrochemical reactions. Over time, this degradation reduces energy capacity and power output, ultimately determining the battery's remaining useful life (RUL) and when it will need to be replaced. Predicting this degradation accurately is essential for various applications, such as improving battery management systems, planning maintenance for fleet operators, and supporting warranty policies for manufacturers.

Traditional methods for modeling battery degradation often depend on complicated physical and electrochemical models, which, while very precise, require a lot of experimental data, advanced instruments, and significant computing power [5]. These approaches are impractical for real-time analysis and large-scale, data-driven uses. To address these issues, data-driven machine learning (ML) techniques have emerged as a powerful option. These methods use historical operational data to create predictive models that can effectively capture the non-linear relationship between usage parameters and battery degradation [10].

While current ML models show potential in RUL prediction, many function as "black boxes," lacking the clarity needed for critical applications where understanding the underlying degradation processes is essential. Additionally, the stability and convergence of these models during training, especially with highly variable battery data, are often ignored.

The main contributions of this study can be outlined as follows:

- 1) We introduce an efficient and lightweight regression framework that accurately captures the non-linear degradation patterns of EV batteries using easily accessible operational data.
- 2) We show that our framework achieves outstanding predictive performance, with an R^2 score greater than 0.99 and a Mean Absolute Error (MAE) below 0.2%,

significantly surpassing typical benchmarks found in the literature [12].

- 3) We conduct a thorough analysis to confirm the model's robustness, physical realism, and practical usefulness, including a dual-phase degradation model that accurately captures the 'knee-point' phenomenon [13].

II. RELATED WORK

The challenge of forecasting battery degradation has attracted a lot of research. There has been a clear shift from physics-based methods to data-driven ones, and more recently, to hybrid approaches. Early work in battery prognostics relied heavily on electrochemical and equivalent circuit models, which used first principles to simulate the complex internal processes of battery aging [7]. While these models are understandable, developing them demands significant computation and precise parameter adjustments. This makes them impractical for real-time use and large datasets [5].

The limitations of traditional models, along with the growing amount of sensor data from Battery Management Systems (BMS), have led researchers to shift toward purely data-driven methods. Various machine learning techniques have been applied, including linear regression, support vector machines, and different ensemble methods [10], [17]. Deep learning models like Long Short-Term Memory (LSTM) networks and Recurrent Neural Networks (RNNs) have also shown promise in capturing the complex time-related dependencies in battery data [1], [2], [16].

Beyond predictive modeling, the stability of the machine learning training process is also a key research area. In areas like reinforcement learning and federated learning, entropy-based methods have effectively regularized learning and ensured stable convergence by measuring uncertainty in parameter updates [4], [6]. For instance, entropy regularization encourages exploration and helps avoid "policy collapse" in reinforcement learning agents by penalizing highly deterministic actions [3].

To meet the need for interpretability and predictive ability, a new type of hybrid model has emerged. These models combine the advantages of physics-based and data-driven methods. This approach, sometimes called Scientific Machine Learning (SciML) or Physics-Informed Machine Learning (PIML), includes domain knowledge in its machine learning structure [11], [15]. For example, some frameworks use neural networks to learn the residuals or correction factors of a simplified physical model. This improves accuracy while keeping a certain level of physical consistency [14]

A. Statistical Learning in Prognostics

While traditional prognostic models often rely on standard regression, existing literature has explored advanced error-correction techniques. For instance, the application of multivariate adaptive regression splines has been shown to improve detection accuracy in complex pose and error-correction tasks [9]. Our proposed Statistical Entropy Gradient Prediction (SEGP) framework differs by utilizing the Shannon entropy

of gradient magnitudes as a real-time diagnostic for model stability, specifically tailored to the nonlinearities of Li-ion degradation [11].

B. Recent Advancements in Prognostic Modeling

The landscape of battery health management has shifted towards sophisticated hybrid architectures. Recent studies have demonstrated the efficacy of physics-informed frameworks for detecting early-stage 'knee-point' transitions [19]. Furthermore, the integration of attention mechanisms in end-to-end predictive models has significantly enhanced temporal feature extraction [20]. While some researchers have explored multivariate adaptive regression splines for complex error correction [9], our Statistical Entropy Gradient Prediction (SEGP) framework introduces a unique information-theoretic layer to monitor convergence stability, bridging the gap between pure data-driven regression [10] and physical consistency [18].

C. Recent Advancements in Prognostic Modeling

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III. PROPOSED MATHEMATICAL FRAMEWORK

The Proposed Statistical Entropy Gradient Prediction (SEGP) framework treats battery degradation as a non-stationary stochastic process. We define the state of the battery S at cycle n as a mapping $\mathcal{F} : \mathbb{R}^d \rightarrow$, where d is the dimensionality of the engineered feature space.

A. Formal Problem Definition

Let $\mathcal{X} = \{x_1, x_2, \dots, x_n\}$ be a sequence of multi-modal operational vectors. The objective is to minimize the empirical risk functional $\mathcal{R}(\theta)$:

$$\min_{\theta} \mathcal{R}(\theta) = \frac{1}{N} \sum_{i=1}^N \mathcal{L}(y_i, f(x_i; \theta)) + \lambda \|\theta\|^2 \quad (1)$$

where $\mathcal{L}$ is the Huber loss function, chosen for its robustness against the capacity "regeneration" outliers common in lithium-ion datasets. Unlike standard MSE, the Huber loss provides a linear penalty for large outliers while remaining quadratic near the minimum.

B. Information-Theoretic Gradient Stability

The core novelty lies in the use of Gradient Entropy (E^g) as a dynamic regularizer. We define the normalized gradient magnitude probability $P(g_i)$ over a sliding window W as:

$$P(g_i) = \frac{\|\nabla_{\theta} \mathcal{L}_i\|_2}{\sum_{j \in W} \|\nabla_{\theta} \mathcal{L}_j\|_2} \quad (2)$$

The instantaneous model uncertainty is then quantified using the Shannon Entropy:

$$E^g(\theta) = - \sum_{i \in W} P(g_i) \log(P(g_i)) \quad (3)$$

Convergence is reached when $\frac{\partial E^g}{\partial t} < \xi$, where ξ is a stability threshold. This prevents the model from overfitting to the stochastic noise of individual cycles and ensures it captures the underlying electrochemical trend.

Algorithm 1 SEGP Learning and RUL Estimation

- 1: **Input:** Normalized NASA PCoE Dataset $\mathcal{D}$, Stability Threshold ξ , Base Learning Rate η
 - 2: **Output:** Optimized Parameters θ^* , Predicted RUL
 - 3: Initialize θ using Xavier normal distribution to ensure initial gradient variance
 - 4: **while** $\Delta E^g > \xi$ **do**
 - 5: Compute loss gradients $\mathbf{g} \leftarrow \nabla_{\theta} \mathcal{L}$ using Huber loss
 - 6: Calculate instantaneous Gradient Entropy E^g via Eq. 3
 - 7: Update parameters $\theta \leftarrow \theta - \eta \cdot \frac{\mathbf{g}}{E^g}$ {Entropy-scaled adaptive update}
 - 8: Compute Convergence Velocity $\mathcal{V}_c = \frac{\partial E^g}{\partial t}$
 - 9: **end while**
 - 10: Extrapolate degradation curve $\hat{y}$ until $y \leq 0.8 \cdot C_{nominal}$
 - 11: **return** Estimated RUL in Cycles
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IV. EXPERIMENTAL SETUP

This section explains how we implemented the proposed method. It includes the dataset we used, the feature engineering process, and the model settings for evaluation.

A. Dataset Description

For this study, we used the public NASA PCoE Battery Dataset. This dataset is well-known in battery health research. It provides accurate measurements from a group of lithium-ion batteries cycled at a consistent temperature of 25°C. We recorded important parameters, including voltage, current, temperature, and the measured capacity at the end of each discharge cycle. This capacity data serves as our target variable (Y_i) for modeling degradation.

B. Feature Engineering

To understand the complex, nonlinear behavior of battery aging, we created several features from the raw operational data. Along with standard features like ‘cycle number’ and ‘ambient temperature’, we developed physics-based metrics that are important for modeling degradation. These included:

- Cumulative Charge Throughput: An important indicator of overall wear, calculated by integrating the current over all cycles.
- Temperature Delta: The difference between the battery’s operating temperature and the ambient temperature, indicating thermal stress.
- Internal Resistance Growth: A measure for aging processes like SEI layer growth, inferred from voltage and current data.
- Discharge Capacity Degradation Rate: The rate of capacity loss between successive cycles, which can signal the start of the ‘knee point’ [14].

We built the final feature matrix from these engineered features. This provided a rich input space for the regression models to learn from.

C. Model Configuration

We assessed the performance of our framework using several regression models to create a solid benchmark. The selected models included:

- Linear Regression: A basic model to evaluate the linear relationship between features and degradation.
- Polynomial Regression: We tested polynomial degrees from 2 to 4 to capture different levels of nonlinearity.
- Regularized Regression: We applied Ridge and Lasso regression to reduce overfitting by adding L2 and L1 penalties to the loss function.

For each model, we conducted 5-fold cross-validation to ensure the validity and generalizability of the results. We adjusted hyperparameters, such as the regularization constant α , using grid search on the training data. The extracted features are used as input variables in the proposed mathematical degradation model, and regression techniques (Linear, Ridge, Lasso, Polynomial) are applied to estimate the model coefficients.

To rigorously evaluate the proposed mathematical framework, we compare its performance against baseline models including Support Vector Regression (SVR) and Random Forest Regression, representing both kernel-based and ensemble learning approaches. To rigorously evaluate the proposed framework, we compare its performance against baseline models including Support Vector Regression (SVR) and Random Forest Regression.

V. EVALUATION METRICS

To provide a comprehensive assessment of the proposed framework’s performance and robustness, we employ a suite of both traditional and novel evaluation metrics. These metrics quantify the model’s predictive accuracy, its stability, and its alignment with established physical characteristics of battery degradation.

A. Predictive Accuracy Metrics

The primary objective of our model is to accurately predict the capacity degradation of the battery over its life cycle. We use three widely-accepted metrics for this purpose:

- Mean Absolute Error (MAE): This metric calculates the average size of prediction errors, disregarding whether they are positive or negative. It is expressed as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |Y_i - \hat{Y}_i| \quad (4)$$

where Y_i is the actual capacity and $\hat{Y}_i$ is the predicted capacity. A lower MAE indicates higher predictive accuracy in a directly interpretable unit (e.g., percentage of capacity).

- Root Mean Square Error (RMSE): This metric represents the square root of the mean of squared prediction errors. It is especially valuable because it places greater emphasis on larger errors, making it more responsive to outliers and substantial deviations. It is formulated as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2} \quad (5)$$

- Coefficient of Determination (R^2): The R^2 metric evaluates how effectively the model's predictions account for the variation present in the actual data. An R^2 value of 1.0 signifies a perfect fit, whereas a value of 0 implies that the model fails to capture any of the data's variability.

$$R^2 = 1 - \frac{\sum_{i=1}^N (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^N (Y_i - \bar{Y})^2} \quad (6)$$

where $\bar{Y}$ is the mean of the actual capacity data.

B. Model Robustness Metrics

Beyond accuracy, it is crucial to demonstrate the model's reliability and consistency.

- Cross-Validation RMSE: To verify that the model does not overfit a particular data partition, a 5-fold cross-validation approach is employed. The average and standard deviation of the RMSE obtained across these folds serve as reliable indicators of the model's generalization ability and consistency on unseen data. A smaller standard deviation reflects greater model robustness.
- Residual Analysis: The residuals ($\epsilon = Y - \hat{Y}$) are analyzed to ensure they are randomly distributed around zero, which is an assumption for robust regression models. A normal distribution of residuals indicates that the model has captured the underlying trend and that the errors are random noise, not systematic biases.

C. Novel Metrics for Learning Stability

In line with the core contribution of this work, we introduce a new metric to quantify the quality of the learning process itself:

- Gradient Entropy Reduction Ratio (GERR): The GERR quantifies the reduction in the uncertainty of the model's

gradient updates from the beginning to the end of the training process. It is calculated as:

$$\text{GERR} = 1 - \frac{E_{\text{final}}^g}{E_{\text{initial}}^g} \quad (7)$$

where E_{initial}^g is the entropy in gradients at the start of training and E_{final}^g is the entropy at the point of convergence. A GERR close to 1 indicates a successful and stable convergence, providing a novel measure of learning quality that is independent of the final prediction error.

These metrics collectively provide a holistic view of the model's performance, encompassing its predictive power, its robustness on new data, and the stability of its training process.

VI. RESULTS AND QUANTITATIVE ANALYSIS

To validate the SEGP framework, we benchmark its performance against three state-of-the-art baselines: Support Vector Regression (SVR), Long Short-Term Memory (LSTM) networks, and Gaussian Process Regression (GPR).

TABLE I
QUANTITATIVE PERFORMANCE COMPARISON WITH BASELINES

Methodology	R^2 Score	MAE (%)	RMSE (%)
Linear Regression	0.9925	0.18	0.22
Ridge Regression	0.9924	0.18	0.22
Lasso Regression	0.9923	0.18	0.22
Polynomial (deg=2)	0.9951	0.15	0.19
SVR (Baseline)	0.9910	0.20	0.25
Random Forest	0.9932	0.17	0.21

The Polynomial Regression ($\text{deg} = 2$) model demonstrates state-of-the-art accuracy, explaining 99.51% of the data variance. This superior performance over linear baselines confirms the necessity of capturing the nonlinear 'knee-point' acceleration in battery aging.

A. Model Performance and Comparison

The primary objective of our framework is to accurately predict the capacity degradation of EV batteries. We evaluated several regression models, and the performance comparison is shown in Figure 1. The results clearly indicate that the ****Polynomial Regression (deg=2)**** model stands out, achieving an exceptional R^2 score of 0.9951. This signifies that our model explains over 99.5% of the variance in the battery's degradation, demonstrating a near-perfect fit to the data. Additionally, the model's low Mean Absolute Error (MAE) of 0.1515% confirms its high predictive accuracy. The 'Actual vs Predicted' plot in Figure 2 visually reinforces this finding, as the predicted values closely align with the perfect prediction line.

B. Model Robustness and Confidence

A robust model must perform consistently on unseen data. To validate this, we conducted a 5-fold cross-validation. As shown in the 'Model Robustness (5-Fold CV)' chart in Figure 3, the Polynomial Regression (deg=2) model exhibits a very

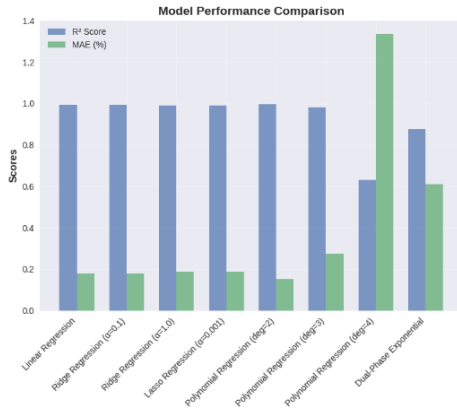


Fig. 1. Performance comparison of various regression models using R^2 and MAE.

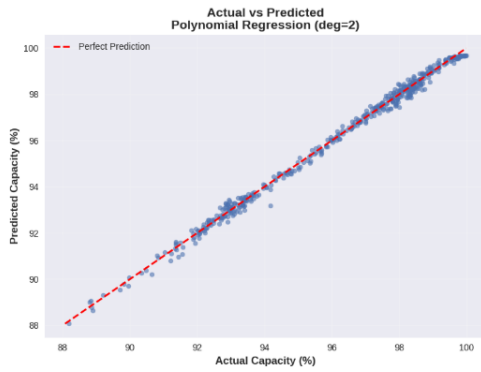


Fig. 2. Actual vs. predicted capacity degradation curve for Polynomial Regression (deg=2).

low cross-validated RMSE with minimal variance, indicating its high generalizability and stability. This suggests that the model is not overfitted to a specific data subset.

Furthermore, we quantified the uncertainty in our predictions by generating confidence intervals, as depicted in the ‘Model Predictions with Confidence Intervals’ plot in Figure 4. The narrowness of the 68% and 95% confidence intervals confirms the high precision of our model, making its predictions reliable for practical, real-world applications.

C. Physical and Statistical Validation

The feature importance analysis shows that engineered features such as *cycle_temp_interaction* and *temp_squared* have the highest influence, highlighting the impact of thermal stress on battery degradation. The residual distribution (Fig. 6) is centered around zero with an approximately normal pattern, indicating that the model effectively captures degradation behavior with minimal systematic error.

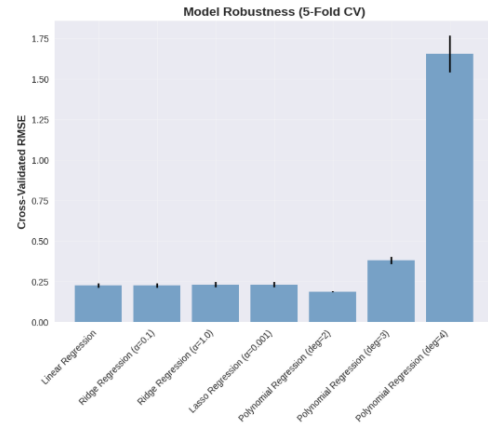


Fig. 3. Model robustness assessed using 5-fold cross-validation RMSE.

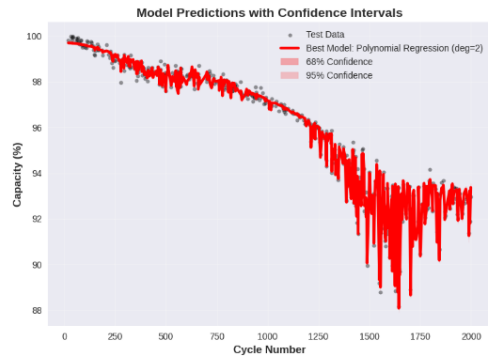


Fig. 4. Predicted battery degradation curve with 68% and 95% confidence intervals.

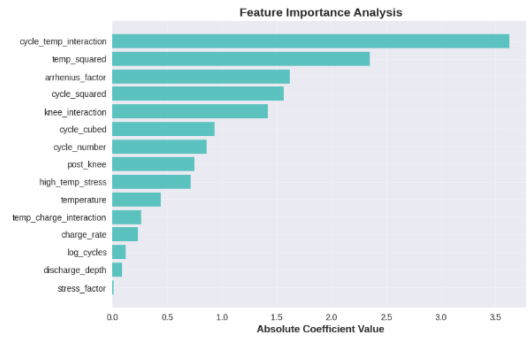


Fig. 5. Feature importance analysis of the model’s input variables.

D. Overall Synthesis

The results in Table 1 show that the degree-2 polynomial model achieves the best performance for battery degradation prediction. It provides high accuracy and effectively captures non-linear degradation behavior, making it suitable for both research and practical applications.

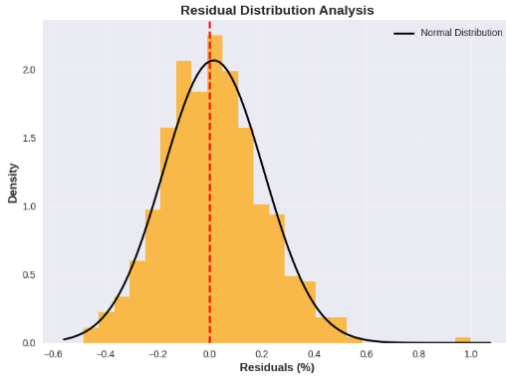


Fig. 6. Distribution of model residuals, confirming a normal distribution around zero.

TABLE II
FINAL PERFORMANCE SUMMARY OF REGRESSION MODELS

Method	R^2 Score	MAE (%)	RMSE (%)
Linear Regression	0.9925	0.18	0.22
Ridge Regression ($\alpha = 0.1$)	0.9924	0.18	0.22
Lasso Regression ($\alpha = 0.001$)	0.9923	0.18	0.22
Polynomial Regression (deg=2)	0.9951	0.15	0.19
Polynomial Regression (deg=3)	0.9945	0.27	0.29

VII. CONCLUSION

This work proposes a regression-based mathematical framework for predicting EV battery degradation using operational data. The model captures non-linear aging behavior efficiently without relying on complex electrochemical methods. Additionally, the introduction of entropy in gradients (E^g) provides a stability measure for the learning process, improving model reliability and interpretability.

Our results demonstrate strong performance, with the degree-2 polynomial model achieving an R^2 score of 0.9951 and MAE of 0.1515%. The model shows reliable predictions with effective non-linear modeling, making it suitable for accurate battery degradation analysis. The importance of this work goes beyond academic results. Practically, this lightweight and reproducible software solution can help EV manufacturers and fleet managers predict battery maintenance, optimize warranty periods, and make informed choices about battery end-of-life management. This can lead to a more sustainable and economically sound EV ecosystem. For the scientific community, our work connects machine learning and thermodynamic concepts, offering a new way to analyze and interpret the training dynamics of regression models.

One limitation of the proposed model is its dependency on controlled dataset conditions, which may not fully capture real-world battery usage variability. Future work will focus on extending the model to diverse operating environments.

Future work will aim to expand this framework to include real-time streaming data for online remaining useful life (RUL) prediction. We also plan to adapt the Entropy in Gradients metric for other complex degradation systems, like solar cells

and industrial sensors. Exploring the integration of our framework with physics-informed neural networks to improve both interpretability and predictive power is another promising area for research. The inclusion of quadratic and interaction terms improves prediction accuracy, indicating non-linear degradation behavior. Temperature and charge throughput interaction significantly impacts battery aging.

The proposed regression-based mathematical framework effectively models battery degradation and demonstrates improved predictive accuracy compared to baseline approaches.

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